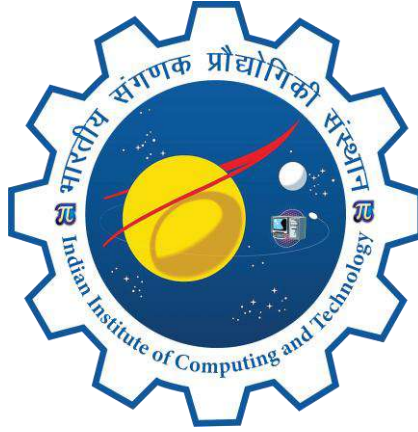


AI- Powered Fake News Detection Using Text Classification



Project Report

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1. Introduction

1.1 Problem Statement

Nowadays, people use the internet and social media platforms to read news every day. News can be shared within a few seconds, but not every news article available online is true. Many fake news articles are created to spread false information, attract attention, or influence people's opinions. Because of this, it has become difficult for users to know whether a news article is real or fake.

Checking every news article manually is not possible because a large amount of news is published every day. Manual verification also takes time and requires reliable sources for fact-checking. As the amount of online news continues to increase, there is a need for a system that can automatically identify fake news with good accuracy.

Machine Learning provides a practical solution to this problem. By learning patterns from previously labeled news articles, a machine learning model can classify new articles as real or fake based on their textual content. Instead of depending only on manual verification, the model can help in making the classification process faster and more efficient.

In this project, a fake news detection system has been developed using Natural Language Processing (NLP) and Machine Learning techniques. The dataset contains both real and fake news articles. After cleaning and preparing the data, different machine learning models are trained and evaluated to identify the model that performs best for fake news classification. The main objective of this project is to build a simple and effective system that can classify news articles using their text.

1.2 Importance of Fake News Detection

Fake news can affect people in many ways. Incorrect information can create confusion, influence public opinion, and spread rumours quickly through online platforms. As more people depend on digital media for daily news, identifying fake information has become an important task.

An automatic fake news detection system can help users by providing a quick way to identify whether a news article is likely to be real or fake. Such systems can also support journalists, researchers, educational institutions, and online platforms by reducing the time required to examine large amounts of news articles.

Natural Language Processing (NLP) plays an important role in fake news detection because machine learning models cannot understand text directly. The text must first be converted into numerical features before it can be used for prediction. In this project, Bag of Words and TF-IDF techniques are used for feature extraction, and multiple machine learning models are trained and compared using the same dataset.

The results of this project show that machine learning techniques can classify news articles with high accuracy when proper data preprocessing, feature extraction, and model evaluation methods are applied. Therefore, fake news detection is an important application of Artificial Intelligence that can help improve the reliability of online information.

2. Dataset Description

2.1 Dataset Source

The dataset used in this project contains two separate files: True.csv and Fake.csv. The True.csv file contains real news articles, while the Fake.csv file contains fake news articles. These datasets were used to build a machine learning model for classifying news articles as real or fake based on their textual content.

The dataset includes news articles from different categories and provides information such as the news title, article text, subject, and publication date. Both files were combined into a single dataset after preparing the data for further analysis and model training.

2.2 Dataset Size

The project uses two datasets that were merged to create a single dataset for machine learning.

Dataset	Number of Records	Number of Columns
True News	21,417	4
Fake News	23,502	4
Total	44,919	4

Table 2.1 Dataset Size

After combining both datasets, a new label column was added to identify the class of each news article. The value 1 represents real news, while 0 represents fake news. The combined dataset was then used for feature extraction, model training, and evaluation.

2.3 Dataset Features

The dataset contains four original features and one target label used for machine learning.

Feature	Description
title	Headline of the news article
text	Complete content of the news article
subject	Category of the news article
date	Publication date of the news article
label	Target variable (1 = Real News, 0 = Fake News)

Table 2.2 Dataset Features

For model training, the title and text columns were combined into a single text field. This combined text was used as the input for feature extraction and machine learning models.

```
saved_news.head()
```

	subject	date	text_length	label	content
0	News	January 1, 2016	2153	0	Obama Announces 'Unfinished Business' For 201...
1	left-news	May 29, 2016	1520	0	IT BEGINS....OBAMA APPOINTED JUDGE Rules Trump U...
2	politicsNews	July 5, 2016	685	1	White House says Obama will not discuss FBI pr...
3	politicsNews	May 12, 2017	1177	1	Treasury unit to share records with Senate for...
4	left-news	Jul 26, 2017	3426	0	HOLLYWOOD LIBS HAVE EPIC MELT DOWNS...Threaten P...

Figure 2.1 Preview of the Combined News Dataset

3. Methodology

3.1 Data Preprocessing

Data preprocessing is an important step in any machine learning project because the quality of the input data directly affects the model's performance. Before training the models, the dataset was prepared to remove unnecessary information and make it suitable for further analysis.

The project started with two separate datasets containing real and fake news articles. The required columns were selected, and the datasets were checked for missing values and duplicate records. After this, both datasets were combined into a single dataset, and a label column was added to identify real and fake news articles. The dataset was then shuffled to avoid any bias during model training.

To prepare the text for machine learning, the title and article text were combined into a single content field. Basic text preprocessing was also performed, including converting all text to lowercase, removing special characters, removing extra spaces, and eliminating common English stop words. These steps helped create a cleaner dataset for feature extraction and model training.



The screenshot displays a Jupyter Notebook interface. The top section shows the command `news.head()` with a green checkmark and a timing of 0.0s. Below this, a table of the first five rows is shown. The columns are `subject`, `date`, `text_length`, `label`, and `content`. The bottom section shows the command `news.info()` with a green checkmark and a timing of 0.0s. Below this, the output of the `info()` method is displayed, showing the dataset's structure and statistics.

	subject	date	text_length	label	content
0	News	January 1, 2016	2153	0	Obama Announces 'Unfinished Business' For 201...
1	left-news	May 29, 2016	1520	0	IT BEGINS...OBAMA APPOINTED JUDGE Rules Trump U...
2	politicsNews	July 5, 2016	685	1	White House says Obama will not discuss FBI pr...
3	politicsNews	May 12, 2017	1177	1	Treasury unit to share records with Senate for...
4	left-news	Jul 26, 2017	3426	0	HOLLYWOOD LIBS HAVE EPIC MELT DOWNS...Threaten P...

```
<class 'pandas.DataFrame'>
RangeIndex: 44919 entries, 0 to 44918
Data columns (total 5 columns):
#   Column      Non-Null Count  Dtype
---  -
0   subject     44898 non-null  str
1   date        44898 non-null  str
2   text_length 44919 non-null  int64
3   label       44919 non-null  int64
4   content     44919 non-null  str
dtypes: int64(2), str(3)
memory usage: 1.7 MB
```

Figure 3.1 Data Preprocessing Steps

3.2 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to understand the dataset before training the machine learning models. This step helped examine the distribution of news articles, identify important patterns, and understand the overall characteristics of the dataset.

Different visualizations were created to study the data. A class distribution chart was used to compare the number of real and fake news articles. Subject-wise analysis was performed to understand the categories available in the dataset. In addition, the distribution of article length was analyzed to observe the variation in the size of news articles.

The insights obtained from EDA helped in understanding the dataset and confirmed that it was suitable for further feature extraction and model training.

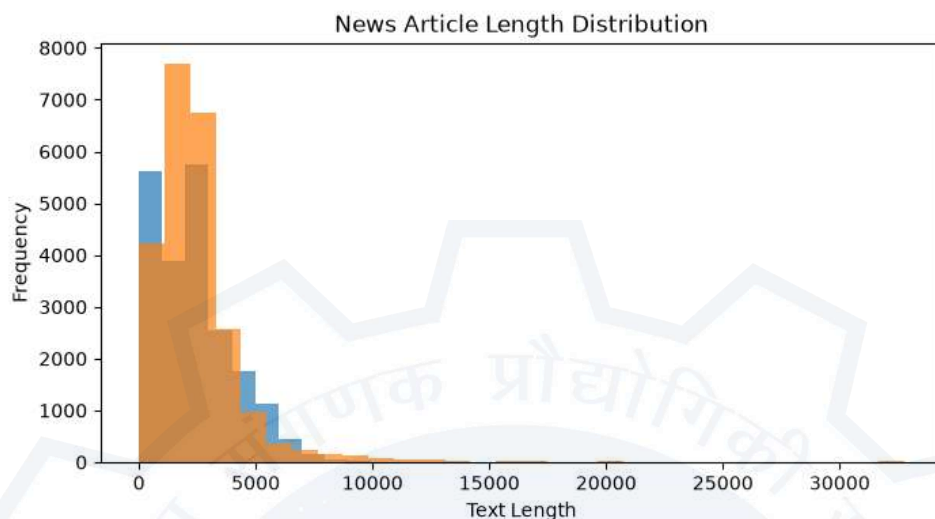


Figure 3.2 Article Length Distribution

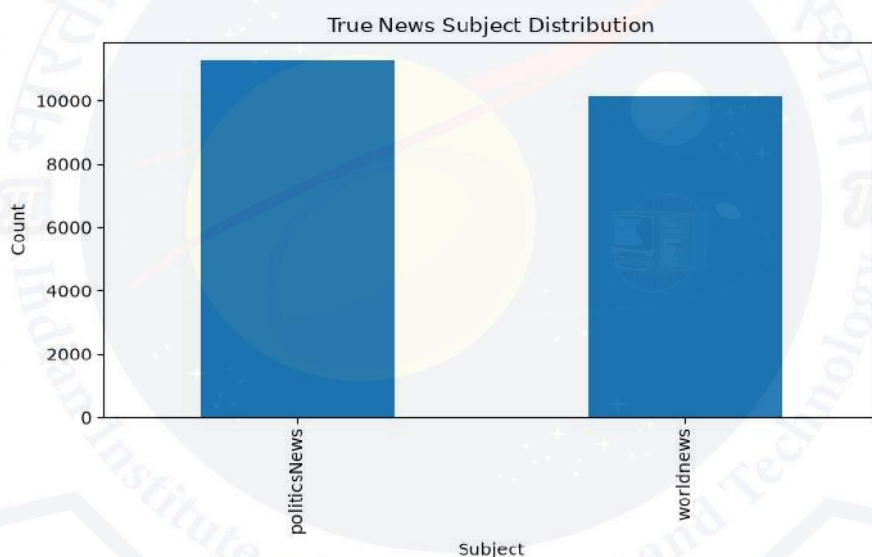


Figure 3.3 Subject-wise Distribution

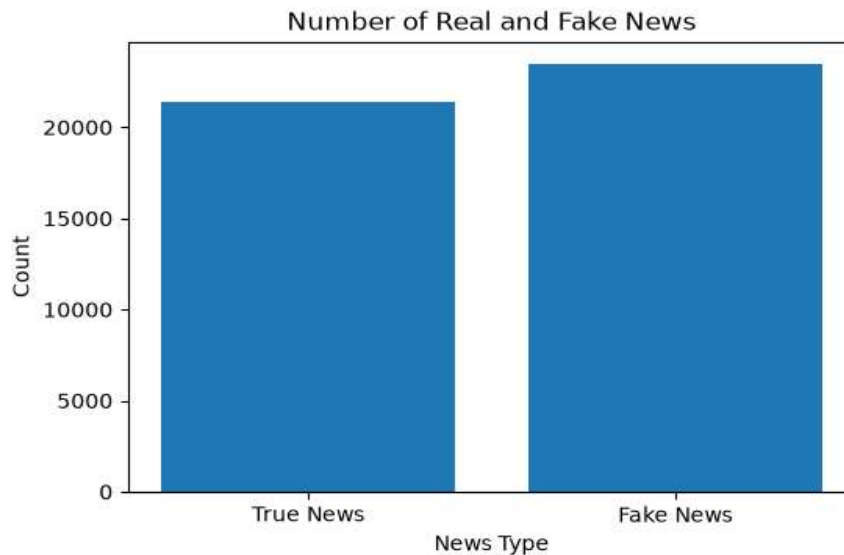


Figure 3.4 Distribution of Real and Fake News

3.3 Feature Extraction

Machine learning algorithms cannot process text directly. Therefore, the textual data was converted into numerical features before training the models.

In this project, two feature extraction techniques were used. The first technique was Bag of Words (BoW), which represents text based on the frequency of words in each document. The second technique was Term Frequency–Inverse Document Frequency (TF-IDF), which assigns weights to words based on their importance in the dataset. TF-IDF reduces the influence of very common words while giving higher importance to meaningful words.

Although both techniques were implemented, TF-IDF was used as the primary feature representation for training and evaluating all machine learning models because it provides a better representation of textual data.

```
news.head()
```

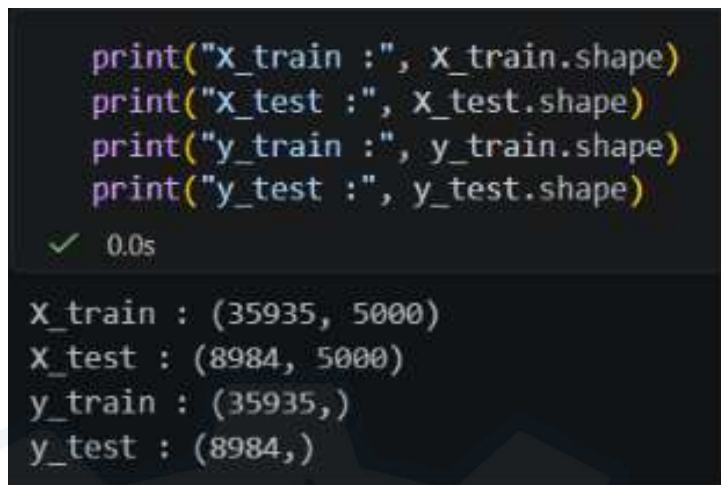
	subject	date	text_length	label	content
0	News	January 1, 2016	2153	0	Obama Announces 'Unfinished Business' For 201...
1	left-news	May 29, 2016	1520	0	IT BEGINS...OBAMA APPOINTED JUDGE Rules Trump U...
2	politicsNews	July 5, 2016	685	1	White House says Obama will not discuss FBI pr...
3	politicsNews	May 12, 2017	1177	1	Treasury unit to share records with Senate for...
4	left-news	Jul 26, 2017	3426	0	HOLLYWOOD LIBS HAVE EPIC MELT DOWNS...Threaten P...

Figure 3.5 TF-IDF Feature Matrix

3.4 Train-Test Split

After feature extraction, the dataset was divided into training and testing sets. The training dataset was used to train the machine learning models, while the testing dataset was used to evaluate their performance on unseen data.

An 80:20 split was used in this project, where 80% of the data was used for training and the remaining 20% was reserved for testing. A fixed random state was used to ensure that the data split remained consistent every time the project was executed.



```
print("x_train :", x_train.shape)
print("x_test :", x_test.shape)
print("y_train :", y_train.shape)
print("y_test :", y_test.shape)

✓ 0.0s

x_train : (35935, 5000)
x_test : (8984, 5000)
y_train : (35935,)
y_test : (8984,)
```

Figure 3.6 Train-Test Split Output

3.5 Machine Learning Algorithms

Five machine learning algorithms were used in this project to classify news articles as real or fake. All models were trained using the same training dataset and tested on the same testing dataset. Using the same dataset and train-test split for every model ensured that the comparison was fair and the performance of each algorithm could be evaluated under the same conditions.

The selected algorithms represent different machine learning approaches and are widely used for text classification tasks. Each model learns patterns from the TF-IDF feature vectors and predicts whether a news article belongs to the real or fake category.

- **Logistic Regression** was used as a linear classification model. It learns the relationship between the input features and the target class and is commonly used for binary classification problems.
- **Naive Bayes** is a probabilistic classification algorithm that is simple, fast, and efficient for text-based datasets. It is widely used in text classification because of its low computational cost.
- **Random Forest** is an ensemble learning algorithm that combines multiple decision trees to improve prediction accuracy. It reduces the risk of overfitting and provides stable performance on large datasets.
- **K-Nearest Neighbors (KNN)** is a non-parametric algorithm that classifies a news article based on the nearest neighbouring samples in the feature space. The prediction depends on the similarity between the new sample and the training data.
- **Neural Network (MLP Classifier)** was used to learn more complex patterns from the TF-IDF feature vectors. It consists of interconnected layers that help the model identify relationships within the textual data.

After training, the performance of all five models was evaluated using accuracy, precision, recall, F1-score, and confusion matrix. These evaluation metrics were used to compare the models and identify the algorithm that achieved the best performance for fake news classification.

4. Results

4.1 Model Performance

After training the machine learning models, their performance was evaluated using the testing dataset. Each model was tested using the same train-test split to ensure a fair comparison. The evaluation was based on standard classification metrics, including accuracy, precision, recall, F1-score, and confusion matrix.

Model	Accuracy (%)
Logistic Regression	98.69
Naive Bayes	92.95
Random Forest	99.51
KNN	88.70
Neural Network	99.25

Table 4.1 Model Accuracy Comparison

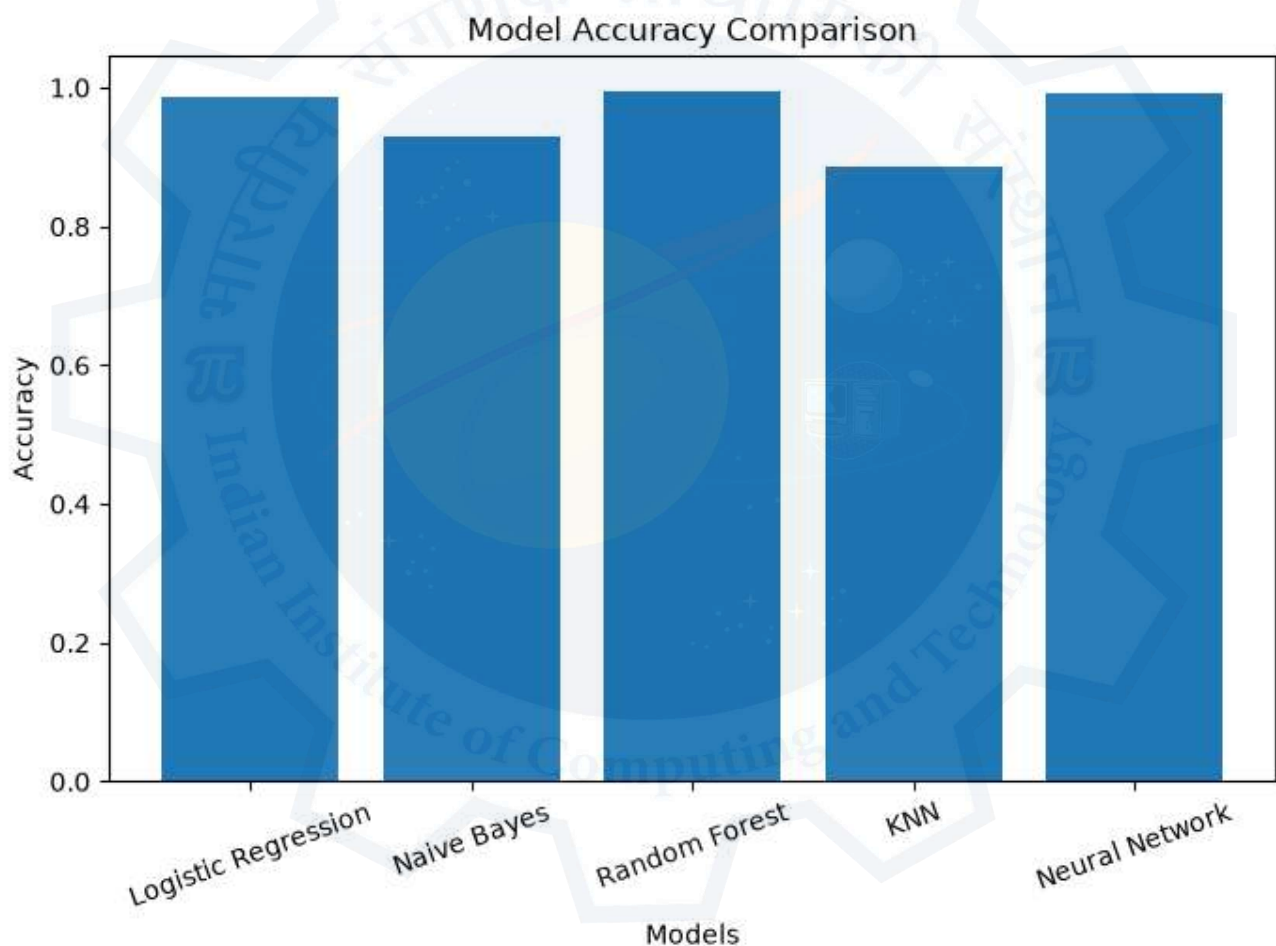


Figure 4.1 Model Accuracy Comparison

The results show that all models achieved good performance on the dataset. However, the accuracy of each model was different because every algorithm uses a different learning approach.

4.2 Classification Report

In addition to accuracy, precision, recall, and F1-score were used to evaluate the classification performance of the models. These metrics provide a better understanding of how correctly each model identifies real and fake news articles.

- **Precision** measures how many predicted news articles are correctly classified.
- **Recall** measures how many actual news articles are correctly identified.
- **F1-score** provides a balanced measure of precision and recall.

These metrics help evaluate the reliability of the machine learning models beyond overall accuracy.

4.3 Confusion Matrix

A confusion matrix was used to visualize the classification results of each machine learning model. It shows the number of correctly classified and incorrectly classified news articles for both real and fake classes.

The diagonal values of the confusion matrix represent correct predictions, while the remaining values represent incorrect predictions. A higher number of correct predictions indicates better model performance.

4.4 Model Comparison

The performance of all five machine learning models was compared using the same evaluation metrics and the same testing dataset. This comparison helped identify the strengths of each algorithm and determine which model performed best for fake news classification.

Among all the models, Random Forest achieved the highest accuracy, followed by the Neural Network and Logistic Regression. Naive Bayes also performed well, while K-Nearest Neighbors (KNN) achieved the lowest accuracy among the five models.

The comparison shows that ensemble and neural network approaches produced better results than the probabilistic and instance-based methods used in this project.

4.5 Best Performing Model

Based on the evaluation results, **Random Forest** was selected as the best-performing model for this project. It achieved an accuracy of **99.51%**, which was the highest among all the machine learning algorithms evaluated.

The model produced high precision, recall, and F1-score, indicating that it classified both real and fake news articles effectively. Its strong performance demonstrates that ensemble learning is well suited for this fake news detection task.

Although the Neural Network and Logistic Regression also achieved excellent results, Random Forest provided the best overall performance on the testing dataset and was selected as the final model for this project.

5. Discussion

5.1 Interpretation of Results

The experimental results show that machine learning techniques can effectively classify news articles as real or fake based on their textual content. All five models achieved good classification performance, although their accuracy varied because each algorithm follows a different learning approach.

Among all the models, Random Forest achieved the highest accuracy and provided the most reliable predictions. Neural Network (MLP Classifier) and Logistic Regression also performed well, showing that they are suitable for text classification tasks. Naive Bayes produced satisfactory results with a simple and efficient approach, while K-Nearest Neighbors (KNN) achieved comparatively lower accuracy on this dataset.

The comparison of different models shows that selecting an appropriate algorithm plays an important role in improving the performance of a fake news detection system. Overall, the results indicate that the models successfully learned meaningful patterns from the news articles and were able to classify them with high accuracy.

5.2 Strengths of the Project

The project has several strengths that contribute to its overall performance and reliability.

- The dataset contains both real and fake news articles, providing balanced data for model training and evaluation.
- Proper data preprocessing improved the quality of the textual data before model training.
- TF-IDF feature extraction helped convert text into meaningful numerical features.
- Multiple machine learning algorithms were implemented and compared using the same dataset and evaluation process.
- Standard evaluation metrics such as accuracy, precision, recall, F1-score, and confusion matrix were used to compare the models.
- The project achieved high classification accuracy, with Random Forest providing the best overall performance.

These strengths demonstrate that the developed system can effectively classify news articles and can serve as a useful example of applying machine learning to text classification problems.

5.3 Limitations of the Project

Although the project achieved excellent results, it also has a few limitations.

- The models were trained using a single dataset and may require additional datasets to improve their ability to generalize to different types of news articles.
- The system classifies news articles based only on textual information and does not verify the factual correctness of the content from external sources.
- The project focuses only on English-language news articles.
- Advanced deep learning models such as LSTM, BERT, or Transformer-based models were not implemented in this project.

- The performance of the system depends on the quality of the training dataset and the preprocessing steps applied before model training.

These limitations provide opportunities for improving the system in future work.

5.4 Future Scope

The performance and usability of the fake news detection system can be improved by extending the project in several ways.

Future work may include using larger and more diverse datasets collected from different online news sources. Deep learning models such as LSTM, BERT, and Transformer-based architectures can also be explored to improve classification performance.

The system can be extended to support multiple languages and integrated into websites, mobile applications, or browser extensions to provide real-time fake news detection. In addition, combining textual analysis with image, video, and social media information may further improve the reliability of the predictions.

These improvements can help develop a more accurate and practical fake news detection system for real-world applications.

6. Conclusion

The objective of this project was to develop a machine learning system that can classify news articles as real or fake using Natural Language Processing (NLP) and text classification techniques. To achieve this objective, the news dataset was collected, preprocessed, converted into numerical features using TF-IDF, and used to train multiple machine learning models.

Five machine learning algorithms, namely Logistic Regression, Naive Bayes, Random Forest, K-Nearest Neighbors (KNN), and Neural Network (MLP Classifier), were implemented and evaluated using the same training and testing datasets. Their performance was compared using standard evaluation metrics, including accuracy, precision, recall, F1-score, and confusion matrix.

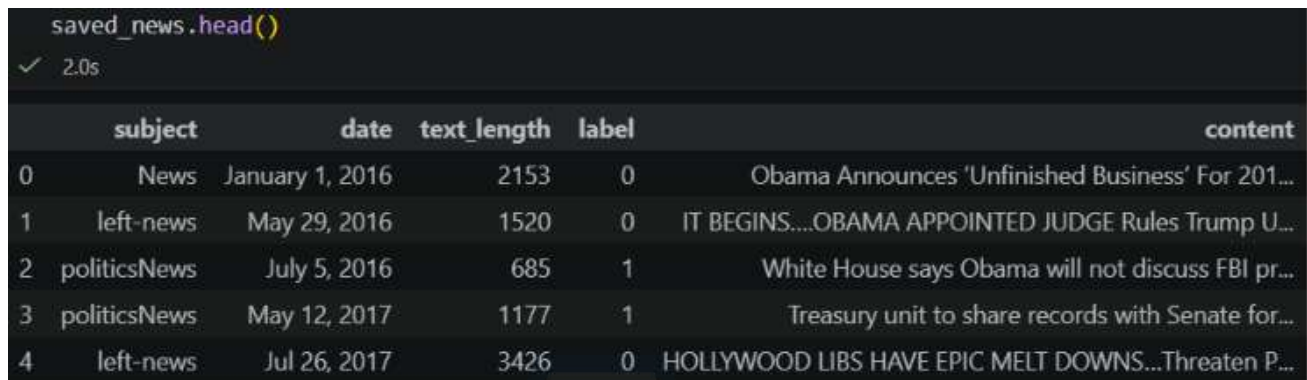
The experimental results showed that all models were able to classify news articles with good performance. Among them, Random Forest achieved the highest accuracy of 99.51%, making it the best-performing model in this project. The comparison of different algorithms demonstrated that machine learning techniques can effectively identify patterns in textual data and accurately distinguish between real and fake news articles.

Overall, this project demonstrates the practical application of machine learning and Natural Language Processing in solving the problem of fake news detection. The developed system provides a simple and effective approach for automated news classification and can serve as a foundation for future improvements and real-world applications.

7. Appendix

The appendix provides supporting information related to the implementation of the fake news detection system. It includes sample data, important code snippets, model outputs, and additional results that support the findings presented in the main report.

Appendix A: Sample Dataset



	subject	date	text_length	label	content
0	News	January 1, 2016	2153	0	Obama Announces 'Unfinished Business' For 201...
1	left-news	May 29, 2016	1520	0	IT BEGINS....OBAMA APPOINTED JUDGE Rules Trump U...
2	politicsNews	July 5, 2016	685	1	White House says Obama will not discuss FBI pr...
3	politicsNews	May 12, 2017	1177	1	Treasury unit to share records with Senate for...
4	left-news	Jul 26, 2017	3426	0	HOLLYWOOD LIBS HAVE EPIC MELT DOWNS...Threaten P...

Figure A.1 Sample of the Merged News Dataset

Appendix B: Important Code Snippets

The following code sections were used during the implementation of the project:

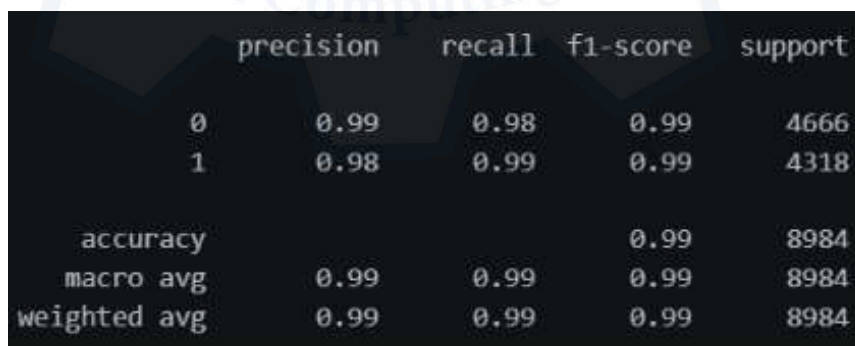
- Import Required Libraries
- Dataset Loading
- Data Cleaning
- Dataset Merging
- Text Preprocessing
- TF-IDF Feature Extraction
- Train-Test Split
- Model Training
- Model Evaluation
- Custom Prediction

Only the key code snippets should be included instead of the complete notebook.

Appendix C: Model Evaluation Outputs

This appendix contains additional outputs generated during model evaluation.

Appendix C contains the classification reports generated for Logistic Regression, Random Forest, and Neural Network models.



	precision	recall	f1-score	support
0	0.99	0.98	0.99	4666
1	0.98	0.99	0.99	4318
accuracy			0.99	8984
macro avg	0.99	0.99	0.99	8984
weighted avg	0.99	0.99	0.99	8984

Figure A.2 Logistic Regression Classification Report

	precision	recall	f1-score	support
0	1.00	1.00	1.00	4666
1	1.00	0.99	0.99	4318
accuracy			1.00	8984
macro avg	1.00	1.00	1.00	8984
weighted avg	1.00	1.00	1.00	8984

Figure A.3 Random Forest Classification Report

	precision	recall	f1-score	support
0	0.99	0.99	0.99	4666
1	0.99	0.99	0.99	4318
accuracy			0.99	8984
macro avg	0.99	0.99	0.99	8984
weighted avg	0.99	0.99	0.99	8984

Figure A.4 Neural Network Classification Report

Appendix D: Confusion Matrices

Appendix D presents the confusion matrices generated for all five machine learning models. These matrices provide a detailed visualization of the classification performance of each model by comparing actual and predicted classes.

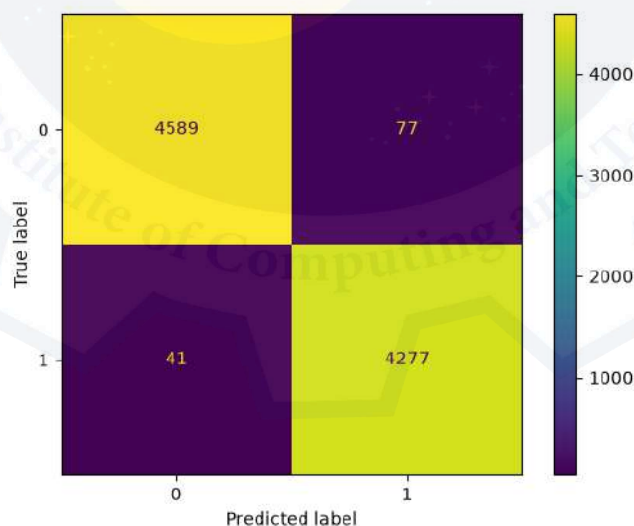


Figure A.5 Logistic Regression Confusion Matrix

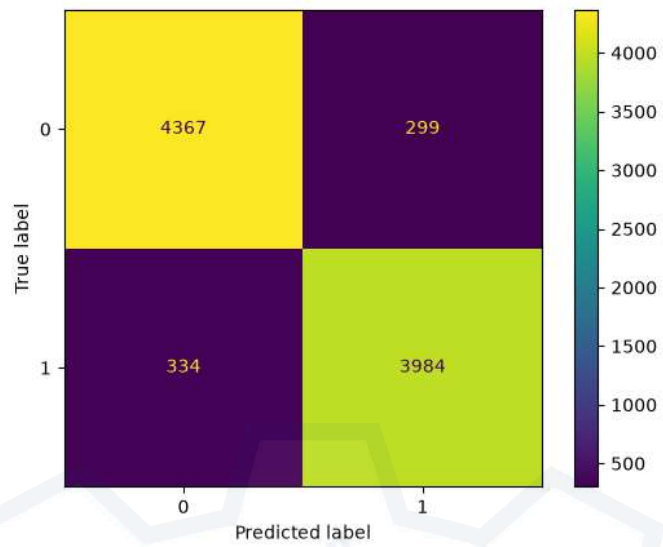


Figure A.6 Naive Bayes Confusion Matrix

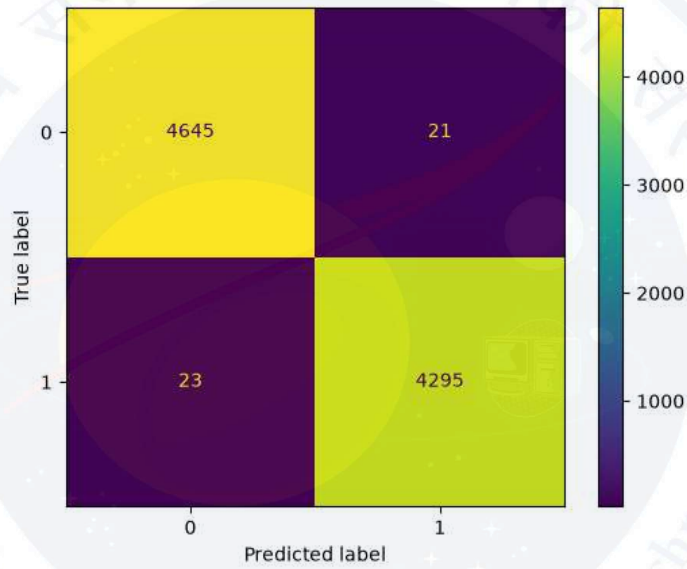


Figure A.7 Random Forest Confusion Matrix

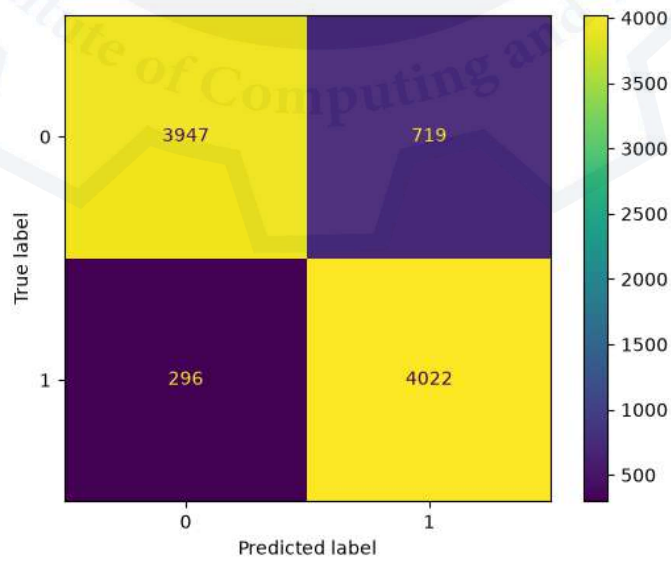


Figure A.8 KNN Confusion Matrix

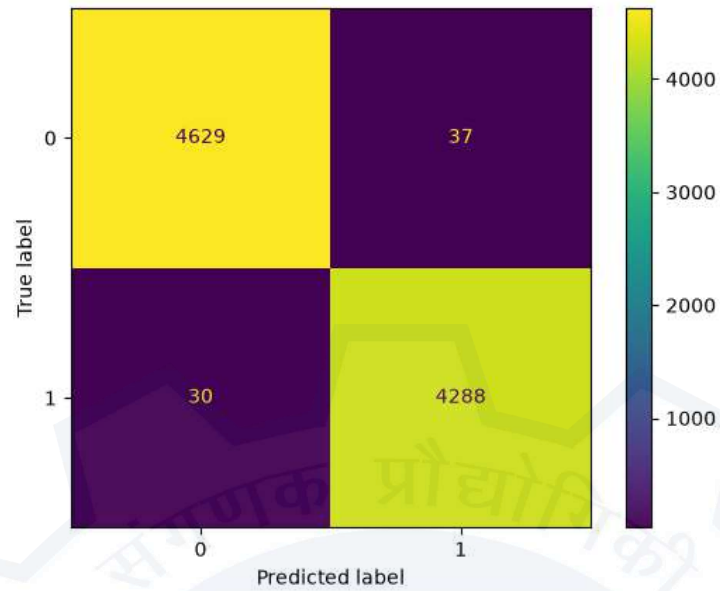


Figure A.9 Neural Network Confusion Matrix

Appendix E: Software Tools and Libraries

The following software tools and Python libraries were used to implement the project.

Software	Purpose
Python	Programming Language
Jupyter Notebook	Development Environment

Table A.2 Software Used

Library	Purpose
pandas	Data Handling
numpy	Numerical Computing
matplotlib	Visualization
re	Text Cleaning
nltk	Stopword Removal
scikit-learn	Machine Learning
joblib	Save Models

Table A.3 Python Libraries

Appendix F: Model Accuracy Summary

	Model	Accuracy
0	Logistic Regression	0.986866
1	Naive Bayes	0.929541
2	Random Forest	0.995102
3	KNN	0.887021
4	Neural Network	0.992542

Table A.4 Model Accuracy Summary

Appendix G: Custom Prediction Examples

The trained model was tested using unseen news sentences to verify its prediction capability.

Input News Sentence	Predicted Output
India launches a new AI education program for students.	Fake News
NASA successfully launched a new satellite into space on Monday.	Real News
The Earth is flat according to new government research.	Fake News

Table A.5 Custom Prediction Results